Charanselvam S

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer — Java, Spring Boot, React — AI Integration — Insurance Domain Expert
Senior Full Stack Engineer with 4.11 years of experience building and maintaining enterprise insurance systems in a regulated homeowners insurance domain. Strong in Java, Spring Boot, Drools BRMS, JSF, and React, with hands-on AI engineering experience including LLM integration, RAG pipelines, and document extraction systems live in production. Actively building AI-powered features and seeking a role where AI engineering is the day job.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL
Frameworks: Spring Boot, React, Next.js, React Native
AI/ML: LLM Integration, Prompt Engineering, RAG, LangChain, ChromaDB, Gemini API, Ollama
Database: PostgreSQL, MySQL
Legacy: Drools BRMS, JSF, XSL, ActiveMQ
Tools: Docker, GitHub Actions, Git, Swagger, Postman, Jenkins, Grafana
Cloud: Oracle Cloud Infrastructure

PROFESSIONAL EXPERIENCE

Solartis Technology Services Pvt. Ltd Chennai
Senior Software Engineer *Jul 2021 – Present*

- Diagnosed and resolved critical ePay payment integration failure affecting 4-5% of homeowner policy transactions, each worth $900-$5,000. Root cause was unhandled exception in JSF Script during transaction ID capture. Implemented defensive exception handling with 10-attempt retry mechanism, reducing silent failures significantly.
- Automated manual operations team renewal workflow using Java, Drools, and MySQL, converting a fully manual process into an automated system, eliminating repetitive ops team work across policy renewal cycles.
- Integrated third-party carrier rating engines and PDF APIs using Drools + JSF workflow, handled Base64 PDF conversion, intelligent split and merge of 70-100 page policy documents using Apache libraries.
- Built JSON-driven dynamic form renderer in React, JSON schema drives entire UI component rendering without code changes. Delivered using GitHub Actions for automated deployment. Live on GitHub
- Automated email ingestion pipelines pushing generated PDFs into TITAN imaging system using JavaMail and Spring Boot, eliminating manual document upload process.
- Resolved long-standing production issue where water damage deductible displayed incorrectly at carrier selection, traced root cause across UI and rating node layers, identifying async data loading timing issue.
- Integrated Concept One, MapRisk, and TITAN third-party systems, including JWT token lifecycle management with automated refresh on expiry for MapRisk API.
- Built AI-powered document extraction service using Google Gemini API which accepts any Excel format regardless of column structure, extracts structured JSON using engineered prompts. Replaced failing IBM WatsonX implementation, reducing parsing failures from 30% to near zero. Containerized with Docker. Live in production. GitHub
- Built PDF RAG chatbot using Python, LangChain, ChromaDB, HuggingFace embeddings, and Llama3, chat with any uploaded PDF using local LLM with hallucination prevention. Live on GitHub.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Generative AI Professional
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional
- Oracle Cloud Infrastructure 2025 AI Foundations Professional
- Oracle Cloud Infrastructure 2025 AI Data Science Professional

EDUCATION

Velammal Institute of Technology
Bachelor of Electronics and Communication Engineering *2017 – 2021*